

Assignment 1: Are Formal Methods Worth It? – A Cost–Benefit Analysis Through a Mini Case Study

You are part of a software consultancy team. A client is developing a software system related to medical device monitoring system for heart patients. The client asks whether formal methods should be used in the development process.

You are supposed to address the following:

System Description: Briefly describe the system's purpose, its critical requirements (e.g., safety, security, correctness) and the impact of failure

Benefits of Using Formal Methods: Identify and explain at least four benefits of applying formal methods to this system.

Costs and Challenges: Identify and explain at least four costs or limitations of this system.

Decision and Justification: Based on your analysis, would you recommend using formal methods? If yes, where (requirements, design, verification)? If no, why not, and what alternatives would you suggest?

Assignment 2: Formal Modeling of an ATM Withdrawal System

An ATM machine allows a customer to withdraw cash from their bank account. The system is sequential (single user, no concurrency).

- a) Briefly describe the ATM withdrawal process, including: Card authentication, Balance checking and Cash withdrawal.
- b) Define the state of the ATM system, including: Account balance, Daily withdrawal limit and Amount withdrawn so far. Also specify the data types and state invariants.
- c) Formally specify the following sequential operations using a formal notation in VDM-SL or Z. Each operation must include: Pre-conditions, Post-conditions and State updates.
 - Authenticate
 - Withdraw
 - EndSession
- d) Explain how each operation's preconditions ensure that the state invariant is preserved.
- e) Provide atleast one invalid scenario and explain why the invalid scenario is rejected by the formal model.

Assignment 3: From Formal Specification to Sequential Implementation

Consider the formal specification of the ATM Withdrawal System that you developed in assignment 2. Implement this system using a sequential programming language, and demonstrate that your implementation conforms to the formal specification by preserving the specified invariants and satisfying the defined pre-conditions and post-conditions.

Assignment 4: Proving the Correctness of a Simple Sequential Procedure

You are given a simple sequential procedure written in a programming language. Your task is to formally prove its correctness with respect to a given specification.

```
Procedure MaxArray(A[0..n-1])
```

```
    max := A[0]
```

```
    i := 1
```

```
    while i < n do
```

```
        if A[i] > max then
```

```
            max := A[i]
```

```
        i := i + 1
```

```
    return max
```

Specification:

Pre-condition: $n > 0$

Post-condition:

- $\max \geq A[j]$ for all $0 \leq j < n$
- $\max \in A$